

April 19, 2026

Professor Wolfgang Kainz
Editor-in-Chief, ISPRS International Journal of Geo-Information

Dear Professor Kainz,

We are pleased to submit our manuscript, “From Where to What: The Geo-intervention Modelling Framework”, for consideration in *IJGI*. Spatially targeted interventions have profound real-world stakes: road traffic crashes alone kill over a million peoples and cost the global economy more than \$3 trillion annually, while tens of millions of preventable deaths from spatially addressable conditions occur worldwide. Recent advances in geospatial AI revealed that location data is predictive of poverty, household income, and long-term health outcomes; Google’s latest Geospatial Reasoning initiative extends these capabilities into public health, crisis response, and climate resilience. Our manuscript contributes a foundational methodology to this space: a framework for generating location-specific interventions from geographic data, moving beyond *where* problems are located, and towards *what* can be done about them.

GIS has excelled at identifying where problems occur, but the application into location-specific actions has received less attention. We propose geo-intervention modelling as an approach to structure this: optimizing actions and their locations against predicted outcomes using geospatial data and machine learning. We demonstrated the approach through a case study on traffic collision reduction in Toronto, Canada, where the framework identified modifications to red light camera, transit shelter, and wayfinding infrastructure predicted to reduce collisions by 5.7%. While transportation served as an example, the framework applies to population health, urban planning, and other domains where location-based decisions affect human outcomes.

We believe this work is both timely and a strong fit for *IJGI*. It speaks directly to the journal’s focus on geospatial artificial intelligence, spatial analysis and decision-making, and spatial data modelling, and connects closely with the journal’s Special Issues on *LLM4GIS* and *Spatial Data Science and Knowledge Discovery*. We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal. All authors have approved the manuscript and agree with its submission to *IJGI*.

We thank you for your consideration and time.

Sincerely,

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On behalf of all co-authors